JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 91, NO. B11, PAGES 11,557–11,570, OCTOBER 10, 1986

Paleomagnetism of Cretaceous and Eocene Strata, San Miguel Island, California, Borderland and the Northward Translation of Baja California

DUANE E. CHAMPION AND DAVID G. HOWELL

U.S. Geological Survey, Menlo Park, California

MONTE MARSHALL

Geology Department, San Diego State University, California

The history of poleward motion of the Baja California peninsula is a subject of continuing geologic investigations, including paleomagnetic studies during the past decade. Geologic correlations seem not to necessitate northward movement, while paleomagnetic data suggest movement of 1000 km or more. We have made paleomagnetic measurements on Cretaceous and Eocene sedimentary rocks of San Miguel Island, California, which are western lithofacies of rocks cropping out on the Baja peninsula. A total of 97 drill cores taken from Upper Cretaceous turbidites, though weakly magnetized, were found to clean best in response to thermal demagnetization, revealing a characteristic remanence. The characteristic magnetization has an inclination 43.9°, declination of 24.0°, and an α_{95} of 3.9°. This direction is 19° shallower and rotated 50° clockwise from the expected direction for the present site position and corresponds to a paleolatitude of 26°N. A total of 117 drill cores of middle Eocene rocks revealed characteristic magnetizations in response to either alternating field or thermal demagnetizations. Some samples suffer from a secondary thermal magnetization from early Miocene volcanic activity, requiring a thermal cleaning that incompletely removes the overprint. A mean direction of magnetization can be derived from the intersection of planes fitted to the paths of demagnetization. The characteristic magnetization for the Eocene rocks has an inclination of 31.9°, a declination of 71.0°, and an α_{95} of 3.8°. This direction is 19° shallower and rotated 79° clockwise from the expected Eocene direction for this site and corresponds to a latitude of 17°N. These paleomagnetic data, in combination with recently published data from the Baja peninsula and Peninsular Ranges of southern California, indicate motion trajectories that mimic the North American craton for the period Cretaceous to late Eocene time. During the Oligocene the Baja peninsula and adjoining regions to the west moved northward relative to the craton, arriving at a pre–Gulf of California position not later than earliest Miocene time. A comparison of the northward motion of the Baja peninsula with that of the Santa Lucia–Orocopia allochthon shows that both areas underwent rapid poleward translations, but these motions occurred at different times.

INTRODUCTION

The history of motion of the Baja California peninsula through geologic time has been the subject of continuing geologic and geophysical investigation. Since the identification of spreading ridges in the Gulf of California, we have understood the peninsula's recent northwestward motion away from Sonora [*Larson et al.*, 1968]. Earlier possible motions, however, have been more difficult to identify, due in part to geologic correlations that can be made between the Baja peninsula and Sonora on rocks of Mesozoic and older age. These correlations suggest, in the main, that no further significant crustal translations are geologically required [*Gastil et al.*, 1981].

Paleomagnetic information from throughout the Western Cordillera contrasts with this apparent geologic stability and suggests significant northward translation and clockwise rotation of crustal fragments (terranes) [*Beck*, 1976, 1980, 1986]. Many of the Mesozoic batholiths of the western Cordillera have participated in this northward motion [*Beck et al.*, 1981]. In particular, early studies of the southern California batholith show it to have been translated northward 11° and rotated 26° clockwise since Cretaceous time [*Teissere and Beck*, 1973]. Subsequent paleomagnetic studies of plutonic and sedimentary rocks from the Baja peninsula have yielded equivalent answers [*Marshall and MacNaboe*, 1979; *Patterson*, 1984; *Hagstrum et al.*, 1985; *Fry et al.*, 1985; *Filmer and Kirschvink*, 1986].

Copyright 1986 by the American Geophysical Union.

Paper number 6B5917.
0148-0227/86/006B-5917$05.00

For this study we made paleomagnetic measurements on Cretaceous and Eocene sedimentary rocks of San Miguel Island that are correlative with Eocene strata of the San Diego and Baja California area (Figure 1). We derived reliable directions of magnetization from the San Miguel Island rocks, and the paleomagnetic results constrain the timing of the separation of the Baja California peninsula and its associated terranes from the North American craton [*Kies and Abbott*, 1983]. Our results also define a latitudinal trajectory for the Baja-Borderland allochthon that can be compared with that of the craton and the trajectory of another southern California composite superterrane, the Santa Lucia–Orocopia allochthon.

GEOLOGY OF SAN MIGUEL ISLAND

San Miguel Island is the westernmost island of the Northern Channel Islands of the California Continental Borderland (Figure 1). The exposed strata are part of the northern flank of an east-west anticlinal ridge along the northern part of the Nicolas terrane [*Blake et al.*, 1982]. The oldest exposed rocks are Upper Cretaceous submarine fan strata, the Jalama Formation of *Weaver et al.* [1969], that crop out along the western and southwestern shores (Figure 2). Conformably overlying these strata are Paleocene-Eocene sediments of the Ferello fan, a submarine fan system restricted to the Nicolas terrane and currently dismembered from its fluvial and paralic source in the San Diego area [*Howell et al.*, 1974; *Howell and Link*, 1979]. The Paleogene strata are either unnamed or informally called the undifferentiated Poso-Canada formations and South Point Sandstone of *Weaver et al.* [1969]. These Pale-

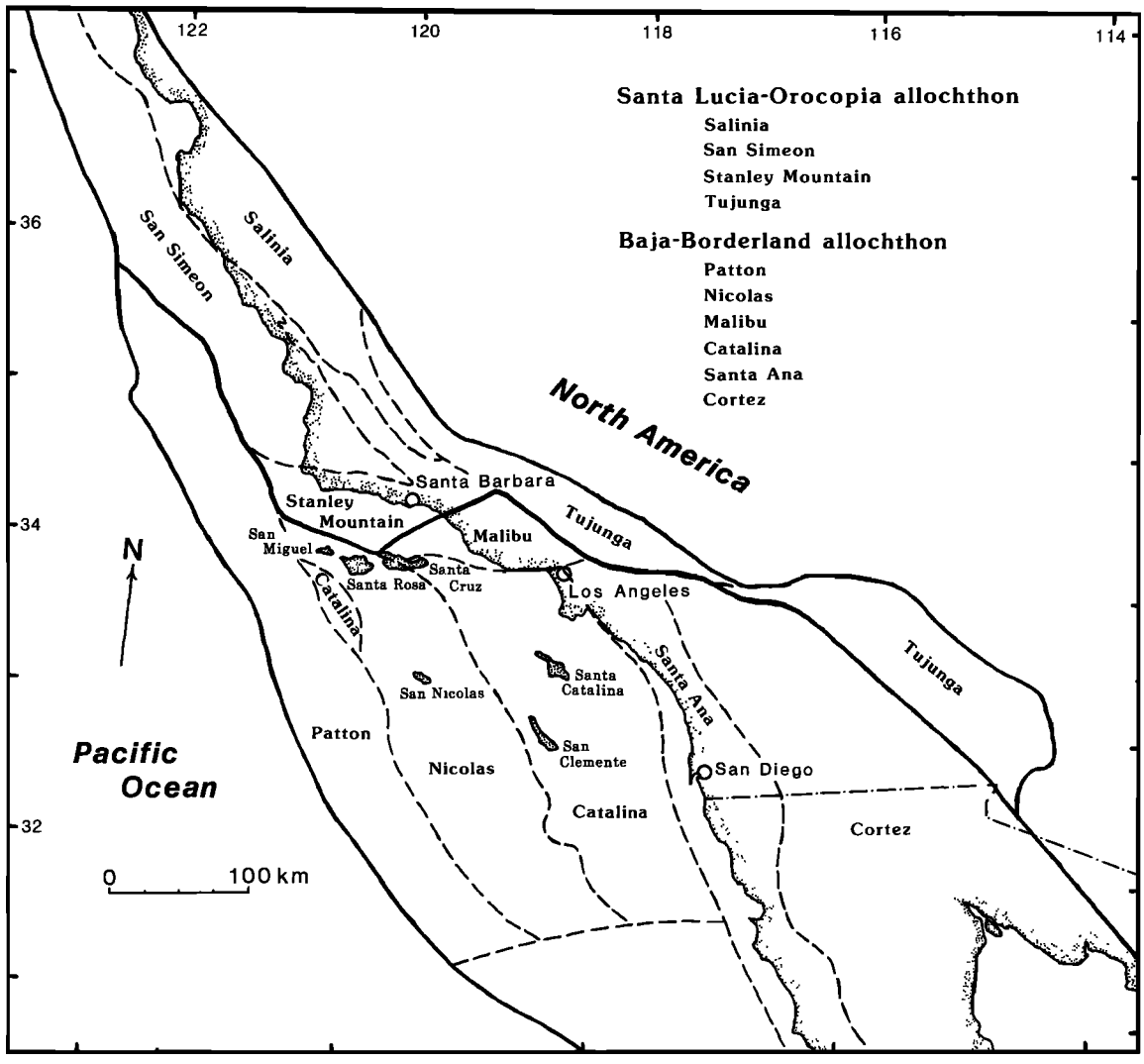


Fig. 1. Terrane map of southern California showing locations of islands in the southern California borderland. Terrane boundaries are those according to *Blake et al.* [1982].

ogene rocks are exposed along the northwestern shores at Simonton Cove, along the north shore in the Cuyler Harbor area, and along the south central portion of the island at Crook Point. Numerous high-angle faults commonly juxtapose the Cretaceous and Paleogene stratal units.

Unconformably overlying these older strata is a relatively thin succession of Oligocene or early Miocene mudstones, the Rincon Formation of *Weaver et al.* [1969], that grades into the basal parts of the overlying San Miguel Volcanics of *Weaver et al.* [1969]. These basaltic and dacitic volcanic rocks, K-Ar dated at about 26 Ma, are best exposed along the northeast shores, particularily at Harris Point. On Prince Island within Cuyler Harbor, the volcanic rock is overlain by thick-bedded, medium-grained sandstone. Elsewhere on San Miguel Island, volcaniclastic units above the volcanic rock grade into sandstone and siliceous mudstone of the Miocene Monterey Formation. Quaternary marine terraces or beach rock cap many shoreline exposures. During the late nineteenth and early twentieth centuries, sheep grazing resulted in devegetation, and most of the island is now covered by eolian deposits that limit rock exposures to beach cliffs and deep gullies. Faults, which as a rule bound all rock units, are mapped variously as normal or reverse, but many may have a strike-slip component. Traces of offshore faults commonly align with those mapped onshore [*Weaver et al.*, 1969], but considering the stratigraphic similarities among rocks of San Miguel, Santa Rosa, and the southern part of Santa Cruz Islands, slip along these faults is difficult to assess and is probably relatively small.

PALEOMAGNETIC PROCEDURES

Core samples for paleomagnetic analysis were taken by using a portable gasoline-powered drill. All cores were oriented with a magnetic compass at the outcrop. Fine-grained hemipelagic mud layers in the upper part of turbidites (Bouma T_e intervals) [*Bouma*, 1962] were preferentially sampled, though occasional coarser-grained horizons (Bouma T_{b-d}) were also drilled. Coarse-grained samples tend to have larger proportions of viscous remanent magnetization within their natural remanent magnetization (NRM) and are thus poorly suited to magnetic cleaning (see *Champion et al.* [1984] for discussion).

The strata directly beneath the Pleistocene deposits typically show a strong orange-red color resulting from oxidation, which can penetrate deeply into coarse-grained layers within the Eocene and Cretaceous strata of San Miguel Island. By not sampling coarse-grained layers we avoid the overprinting magnetizations of the authigenic iron-oxide minerals, which

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11557 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

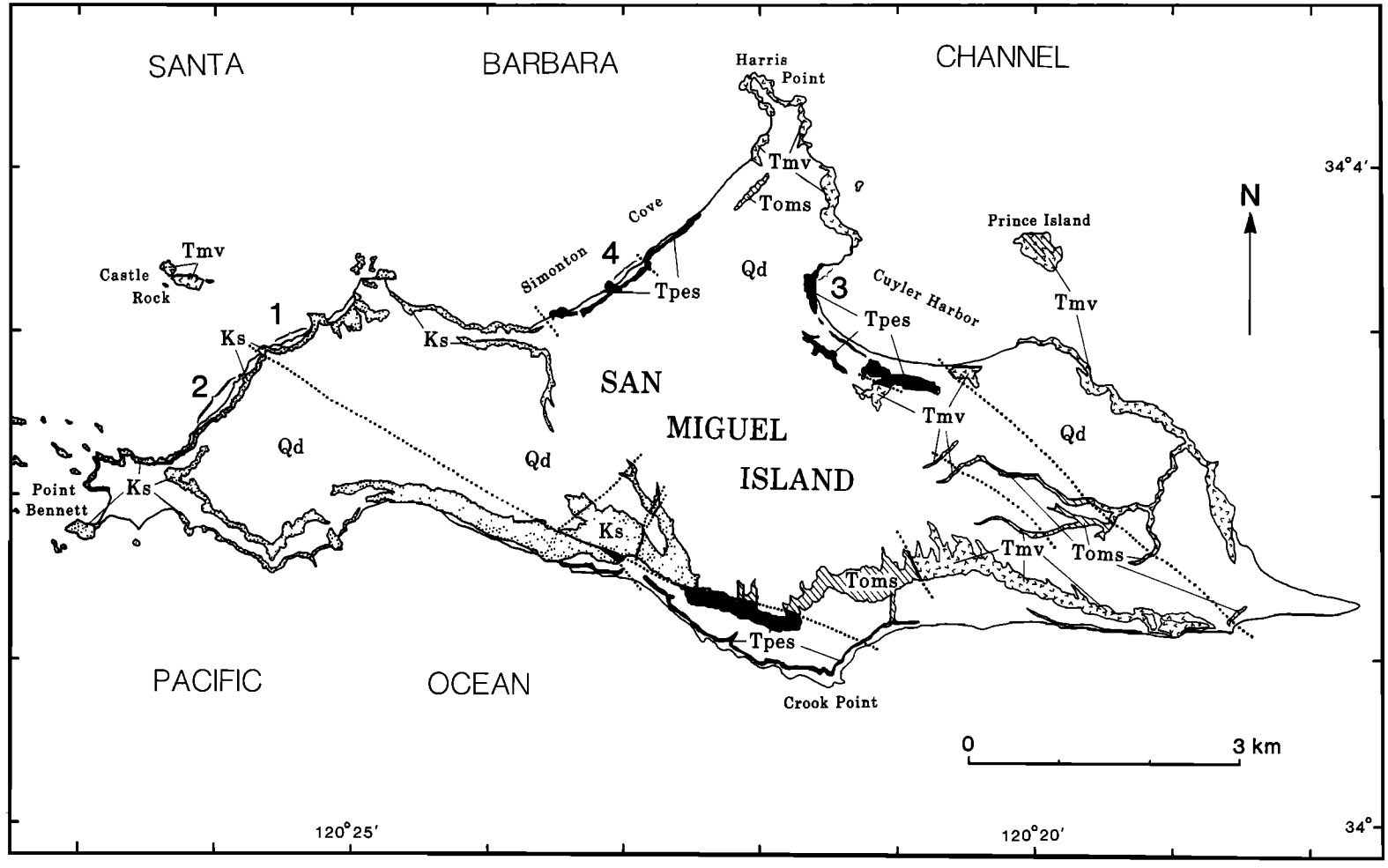


Fig. 2. Generalized geologic map of San Miguel Island, California, showing numbered paleomagnetic sampling localities (see Tables 1 and 2), major faults (dotted lines), and principal geologic units modified from *Weaver et al.* [1969] and R. P. Kies and W. A. Bartling (written communication, 1982). Ks, Upper Cretaceous sedimentary rocks; Tpes, sedimentary rocks of Paleocene-Eocene age; Toms, sedimentary rocks of Oligocene-Miocene age; Tmv, Miocene volcanic rocks; and Qd, Quaternary deposits.

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11557 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

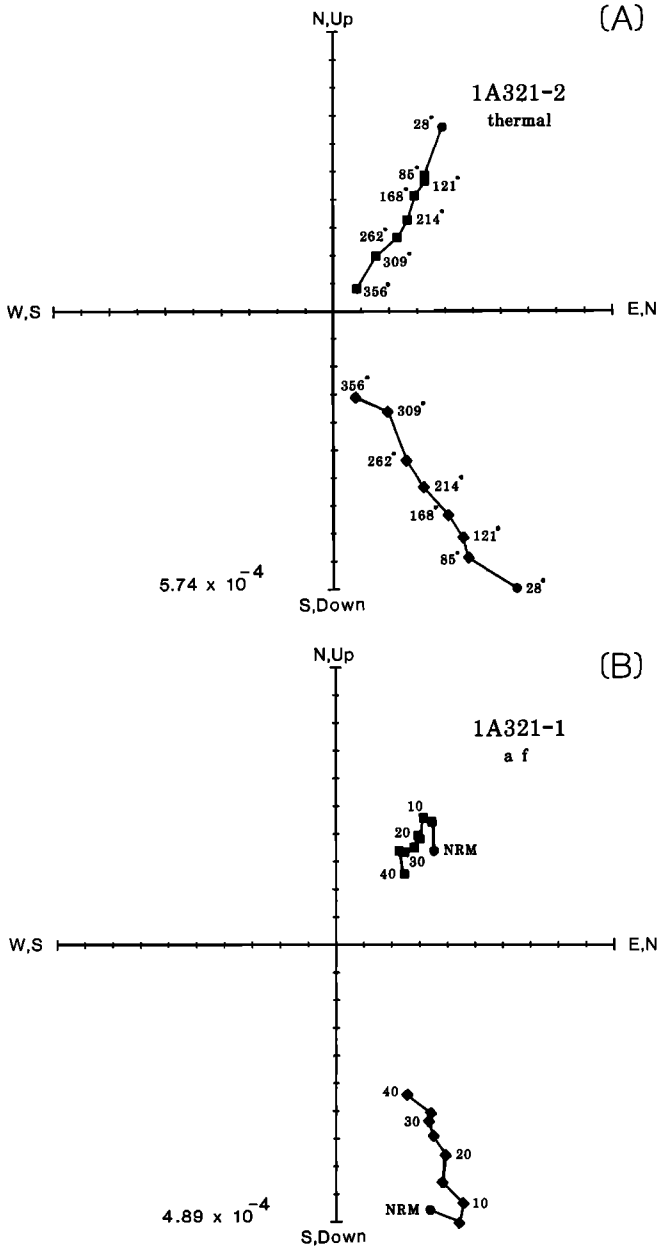


Fig. 3. Orthogonal vector component diagrams of thermal and alternating field demagnetizations of two specimens from core 1A321 from Upper Cretaceous strata near Point Bennett, San Miguel Island. Squares denote the projection of the horizontal components, and diamonds denote the projection of the vertical components. Intensity values at base of diagrams indicate length of component axes in units of amperes per meter: (*a*) specimen 2, showing thermal cleaning at indicated temperatures producing linear descent toward the origin and (*b*) specimen 1, showing alternating field cleaning at indicated peak fields (in milliteslas) producing a less well defined linear descent toward the origin.

usually obscure the original magnetic direction of a sample. Bedding attitudes were measured at frequent intervals during the sampling in order to correct for minor variation of tectonic tilt. The 2.5-cm-diameter cores were cut in the laboratory into specimens 2.2 cm long. The direction and intensity of NRM was measured in one specimen from each core, and standard stepwise demagnetization techniques were followed.

PALEOMAGNETIC RESULTS FROM CRETACEOUS STRATA

Paleomagnetic collections were made at two locations along the west coast of San Miguel Island (Figure 2): site 1 was southeast of Castle Rock, an offshore islet, and site 2 was located more to the southwest, in an area locally known as "Bud's Bathhouse." The strata at both sites are Upper Cretaceous turbidites of a submarine fan system. Drilling targets were fine-grained, thin-bedded overbank-levee facies [*Bartling*, 1981] that are best suited for paleomagnetic investigation (see *Champion et al.* [1984] for rationale). Fossils in the Cretaceous rocks indicate Turonian to late Maastrichtian ages (91–65 Ma); however, we are unable to define the precise age of the beds sampled [*Bartling*, 1981]. Their normal polarity, however, suggests a pre-Campanian age (91–83 Ma).

A total of 97 cores were taken at approximately 1-m intervals from the two sampling sites, 46 from site 1 and 51 from site 2. The beds at the two sites typically strike between 280° and 300° and dip from 20° to 35°, though bedding attitudes are more variable near faults which cut the section.

The NRM intensities for these core samples are low (7×10^{-4} to 1×10^{-3} A/m) but well within the measuring range of a cryogenic magnetometer. Stepwise alternating field (af) demagnetization on a number of core samples and thermal demagnetization on second specimens from the same cores enable a comparison of the two cleaning techniques. Low coercivities and generally low blocking temperatures suggest that magnetite is the principal carrier of remanence in these rocks. Vector component diagrams demonstrate that thermal cleaning better isolates the primary component, producing horizontal and vertical components, which linearly descend toward the origin (Figure 3). Because of this advantage, thermal demagnetization for these samples was selected as the appropriate cleaning technique. However, a direct comparison of the best cleaned directions of magnetization obtained from af and thermal cleaning of 25 specimen pairs from the same number of original cores shows that the mean directions are not significantly different at the 95% confidence level (Figure 4).

Following stepwise thermal cleaning on each specimen, re-

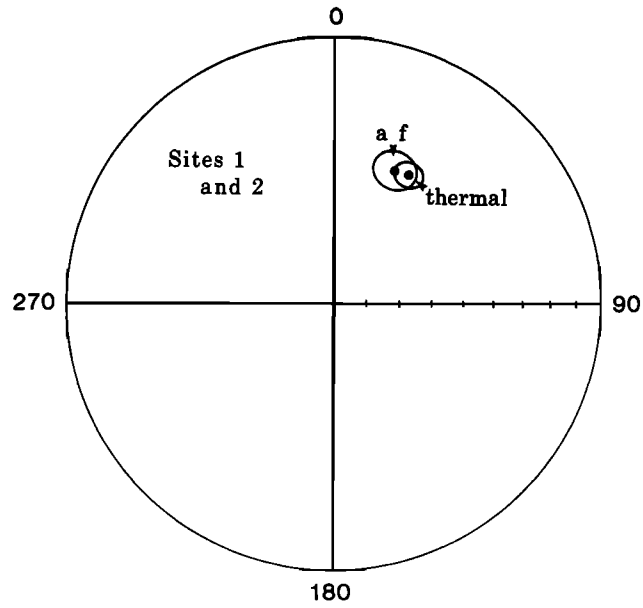


Fig. 4. Equal-area plot of structurally corrected, thermal and alternating-field-cleaned paleomagnetic directions with 95% confidence circles from 25 core samples taken at sites 1 and 2 in Upper Cretaceous strata of San Miguel Island. Despite poorer cleaning results following alternating field techniques the resultant mean field direction is identical at the 95% confidence level to the mean field direction of specimens from the same 25 cores cleaned by thermal demagnetization.

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11557 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

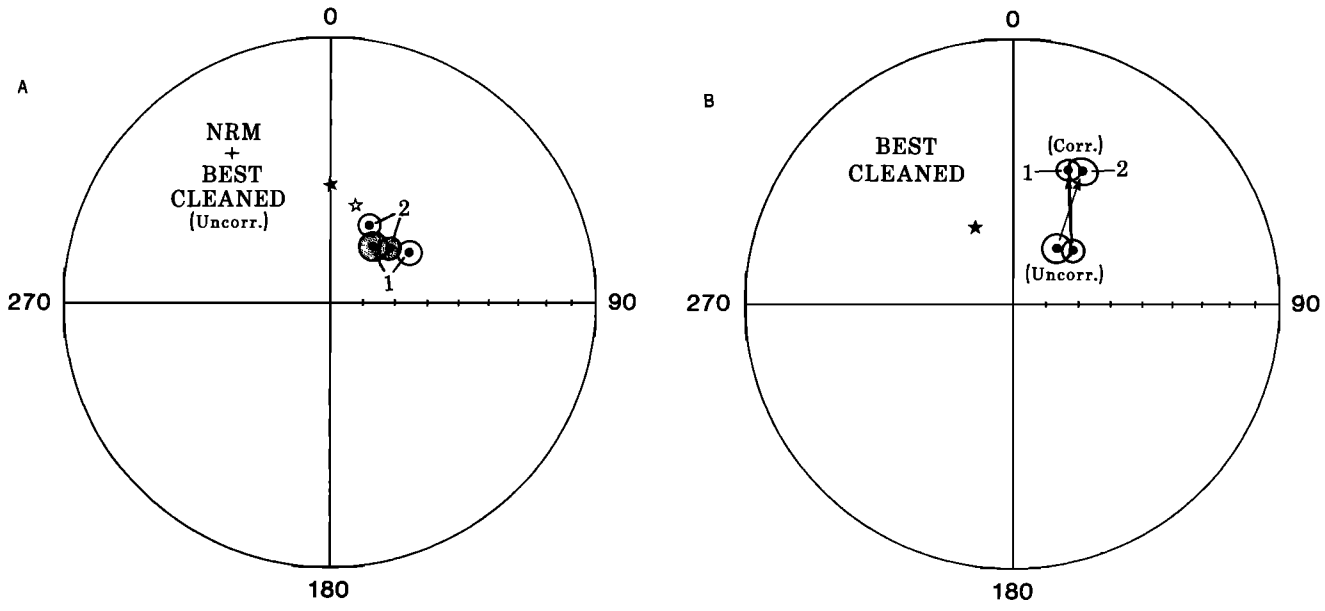


Fig. 5. (*a*) Equal-area plot of structually uncorrected NRM and thermally best cleaned field directions shown with 95% confidence circles from sites 1 and 2 in Upper Cretaceous strata, San Miguel Island. The solid star shows the present axial dipole direction, and the open star shows the present (1980) field direction for the collecting sites. The stippled confidence circles represent the best cleaned thermal data, and the open confidence circles represent the NRM data. (*b*) Equal-area plot of thermally best cleaned field directions, with 95% confidence circles, from sites 1 and 2, showing both structurally uncorrected and corrected results. The star denotes the expected Upper Cretaceous field direction for normal polarity at the position of San Miguel Island, assuming continguity with North American craton. The site mean directions are identical at the 95% level, whether they are structurally corrected or not, but the probability that they are different decreases with correction.

sults from 94 of 97 cores were averaged together by site. Of these, 85 samples have vector component diagrams with linear segments that trend toward the origin during the demagnetization procedure (Figure 3). Nine samples show good cleaning behavior and stable magnetic directions, but they do not produce linear segments on vector component diagrams. We rejected three samples because of anomalous directions of magnetization or irregular cleaning behavior. Samples from sites 1 and 2 differ in their response to thermal demagnetization. An equal-area plot of site means for structurally uncorrected NRM and demagnetized data shows that samples from site 2 have a secondary component near the present field at the site, while samples from site 1 have a component which is shallower in inclination and more easterly in declination (Figure 5*a*). The demagnetization behavior of samples from site 2 is typical of that for thin-bedded turbidites from fresh shoreline exposures. Viscous remanent magnetic effects occur in the coarser grains of fine sand and siltstone samples and are easily removed by weak alternating fields or low-temperature thermal demagnetization. The secondary component removed from site 1 is probably due to partial thermal remagnetization effects from nearby Miocene volcanics and is more fully explained in the forthcoming section dealing with Eocene paleomagnetic results.

Fold tests are difficult to perform on paleomagnetic data from San Miguel Island because of the homoclinal nature of the strata. Nonetheless, a limited tilt test on demagnetized data before and after structural correction demonstrates a tighter clustering of the data after structural correction. Disperson for site 1 decreased with structural correction, producing a small improvement in the precision parameter k from 23.8 to 25.2 (Table 1). For site 2, which has a greater diversity of bedding attitudes, structural correction improved the k from 33.4 to 41.7. Though this degree of improvement is substantial, it is not significant at the 95% confidence level. Similarly, the comparison of mean directions of demagnetized data for sites 1 and 2, before and after structural correction, shows that correction improves the match of mean directions, but not significantly at the 95% confidence level (Figure 5*b*). In particular, the probability that the directions are different decreases from 91% to 83% [*McFadden and Jones*, 1981].

The paleomagnetic results from the two sampling areas of Upper Cretaceous strata on San Miguel Island can be averaged together in a number of ways (Table 1). If only $N = 2$ sites are considered, the k is large, but the confidence in the mean (α_{95}) is low. Similarly, if we take $N = 94$ cores, the k becomes a more realistic 31.7, but we probably overestimate our precision in the mean direction. Neither method of averaging influences the mean direction, which retains incli-

TABLE 1. Paleomagnetic Data After Thermal Demagnetization, Structurally Uncorrected and Corrected, Taken From Sites 1 and 2 in Cretaceous Strata, San Miguel Island

Site	Data	N/N_0	R	k	α_{95}	I	D
1	uncorr	44/46	42.18959	23.8	4.5	68.5	37.0
	corr		42.29469	25.2	4.4	43.1	26.7
2	uncorr	50/51	48.53246	33.4	3.5	65.6	47.5
	corr		48.82390	41.7	3.2	44.6	21.5
Outcrop areas		2/2	1.99875	797.3	8.9	43.9	24.2
Independent cores		94/97	91.06173	31.7	2.6	43.9	24.0
Independent orientation		41	...	...	3.9	...	...

Two data sets, one structurally corrected (corr) and the other uncorrected (uncorr) corresponding to each of two outcrop areas. N/N_0, number of independently oriented cores used in averages relative to the number of cores originally taken; R, length of the resultant vector; k, Fisherian dispersion parameter; α_{95}, radius of the cone of 95% confidence about the mean direction; I, inclination in degrees downward; and D, declination in degrees east of north. Site numbers correspond to those in Figures 2, 4, and 5.

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11557 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

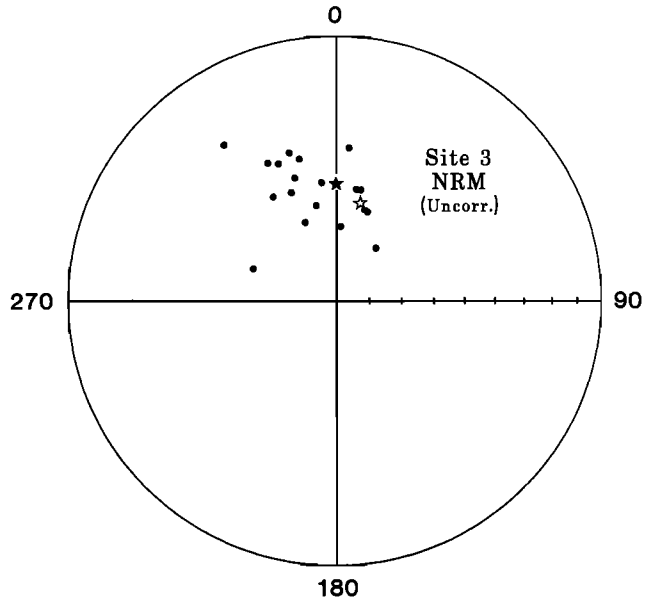


Fig. 6. Equal-area plot of structurally uncorrected NRM directions from site 3, in Eocene strata of Cuyler Harbor. The solid star shows the present axial dipole direction, and the open star shows the present (1980) field direction for the collecting site.

nation values of 43.9° and declination values of 24.1° ± 0.1°. A better figure for a 95% confidence limit can be calculated for the $N = 41$ independent structural attitudes, using the $k = 31.7$ value and giving a value of $\alpha_{95} = 3.9°$.

The mean direction of magnetization of the Cretaceous strata is quite different than that expected for San Miguel Island's present position with respect to cratonal North America in Late Cretaceous time [*Hagstrum et al.*, 1985]. It is 19.6 ± 3.7° shallower and has rotated 50.0° ± 5.7° clockwise from the expected direction. The mean inclination value corresponds to a paleolatitude of 25.7°N, and the inclination anomaly indicates an apparent poleward motion of 19.3°. This paleolatitude corresponds to a present location just south of the mouth of the Gulf of California.

PALEOMAGNETIC RESULTS FROM EOCENE STRATA AT CUYLER HARBOR

Samples from site 3, at the Cuyler Harbor area on the north coast of San Miguel Island, are in Eocene strata (Figure 2). Fresh unweathered strata crop out only at the northwest part of the harbor where the exposure is wave washed and well exposed but close to overlying Miocene volcanic rocks. The Eocene strata are submarine fan lithofacies [*Kies*, 1982]. Micropaleontological evidence at site 3 indicates a middle Eocene age, provincial Ulatisian to Narizian Stages (roughly 52–42 Ma).

A total of 21 cores were collected at approximately 3-m intervals from beds striking 270°–290°, with most dips 32°–51°. In the vicinity of faults, attitudes are more variable, with 2 samples having 10° and 12° dips.

The NRM intensities for these core samples are also low but somewhat stronger than the Upper Cretaceous samples; many have values between 1×10^{-3} and 1.5×10^{-3} A/m. The NRM directions for site 3, uncorrected for structure, are grouped near the direction of the present field (Figure 6). Stepwise af demagnetization results in less than ideal behavior. Generally, directions move along two-component paths toward a reversed polarity direction with westerly declination, but the results become erratic far from their primary direction (Figure 7). Some samples at high-peak fields retrograde and return toward the present field direction. Thermal demagnetization was performed on companion specimens from 20 of 21 original core samples; one core produced only a single specimen. Vector component diagrams demonstrate that thermal demagnetization better removes secondary components in these samples and produces longer two-component cleaning paths than af demagnetization does (Figure 7). In no case, however, did thermal cleaning produce vector component diagrams with linear segments that trend toward the origin, or stable end point directions. Instead, directions move systematically throughout the thermal demagnetization process until chaotic directions appear at high temperatures.

Since none of the samples yielded a stable direction after af or thermal demagnetization, a mean direction of magnetization was calculated from the distribution of poles normal to the demagnetization planes produced by the thermal cleaning [*Kirschvink*, 1980]. The results from 19 of 21 original cores were averaged together for site 3. One sample was rejected due to an anomalous direction of magnetization. The angular path lengths of vector motions during thermal

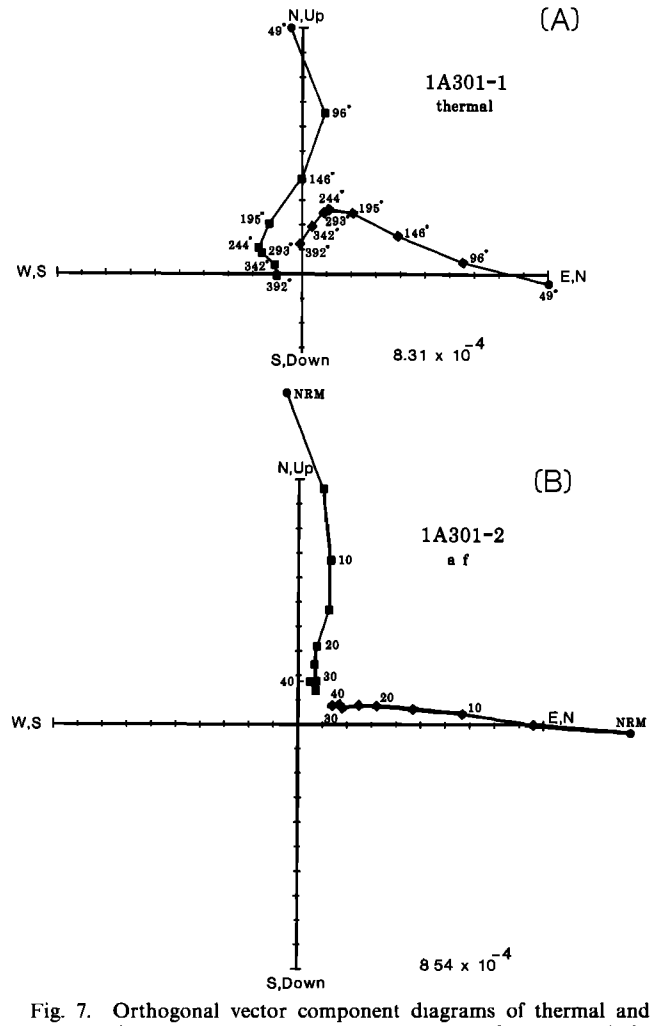


Fig. 7. Orthogonal vector component diagrams of thermal and alternating field demagnetizations of two specimens from core 1A301 from Eocene strata at Cuyler Harbor. Squares denote the projection of the horizontal components, and diamonds denote the projection of the vertical components. Intensity values at base of diagrams indicate length of component axes in units of amperes per meter: (*a*) specimen 1, showing thermal demagnetization behavior at indicated temperatures, and (*b*) specimen 2, showing alternating field demagnetization behavior at indicated peak fields (in milliteslas). Thermal demagnetization better defines the directional components within cores from this site.

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11557 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

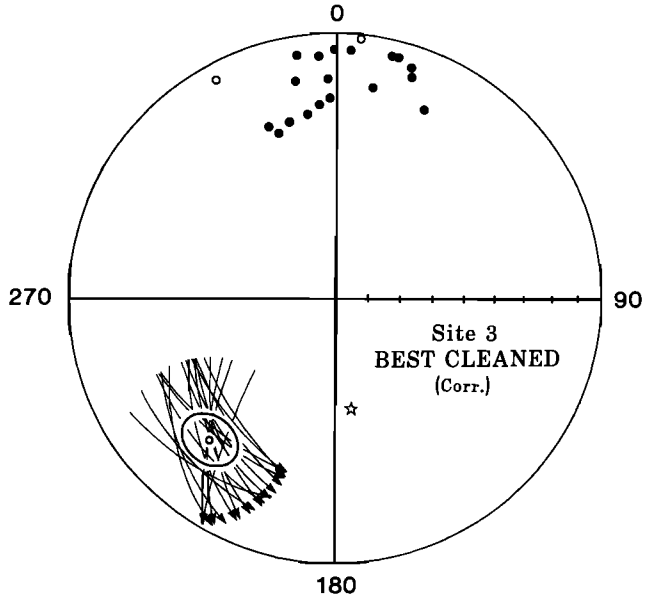


Fig. 8. Equal-area plot of structurally corrected NRM field directions from site 3, Eocene strata of Cuyler Harbor. Arcs with arrows denote the trace of great circle thermal-cleaning paths from these NRM directions and the sense of the cleaning. Mean direction of magnetization with 95% confidence circle is calculated from the distribution of poles to these planes by the method described by *Kirschvink* [1980]. Open star denotes the expected Eocene field direction for reversed polarity at the position of San Miguel Island, assuming continguity with North American craton.

demagnetization for samples from Cuyler Harbor vary from 35° to 131°, and the mean angular path length is 98°. An equal-area plot of the intersection of the demagnetization planes, compared with structurally corrected NRM directions, shows that the path lengths are 1/2 to 2/3 of the angular distance between these points (Figure 8). As such, the intersection of these planes is well constrained, particularly in inclination. The larger of the two dispersion parameters using Bingham statistics [*Kirschvink*, 1980] is quite reasonable at 37.1, and the 95% confidence limit in the mean direction is 8.2°, calculated from the rms of the two orthogonal α_{95}, again using Bingham statistics. The mean direction of magnetization has an inclination of $-29.4°$ and a declination of 221.3°. This direction is $26.4° \pm 6.8°$ shallower and rotated $50.0° \pm 7.9°$ clockwise from the expected Eocene reversed polarity direction. The mean inclination value corresponds to a paleolatitude of 15.7°, and the inclination anomaly indicates an apparent poleward motion of 20.6° with respect to North America.

PALEOMAGNETIC RESULTS FROM EOCENE STRATA AT SIMONTON COVE

A second collection of Eocene strata came from the northwest shore of San Miguel Island at Simonton Cove (site 4, Figure 2). The strata are well exposed and wave washed, and few areas are covered or display weathering oxidation. On the basis of microfossils, the submarine fan strata of this section are early Eocene, provincial Penutian Stage (52–55 Ma).

A total of 96 cores, taken at approximately 1-m stratigraphic intervals, came from beds striking 300°–320° and dipping from 25° to 40°. A break in the sampling, after the first 34 cores, reflects a covered interval several tens of meters wide. An abrupt lithofacies change across this covered section may indicate fault displacement [*Kies*, 1982]. Because of this and numerous other small faults, we are unable to reconstruct the exact stratigraphic thickness, although *Kies* [1982] estimates it to be 600 m.

The NRM intensities for the site 4 cores are low and fall into two groups that correspond to the two sides of the covered interval. Intensities for samples stratigraphically below the covered interval are stronger, 6×10^{-4} to 3×10^{-3} A/m, while those for samples stratigraphically above the interval are weaker, 2×10^{-4} to 1×10^{-3} A/m. The directions of NRM for site 4, uncorrected for structure, also reflect this grouping (Figure 9). Samples from below the covered interval cluster away from the present field direction and have easterly decli-

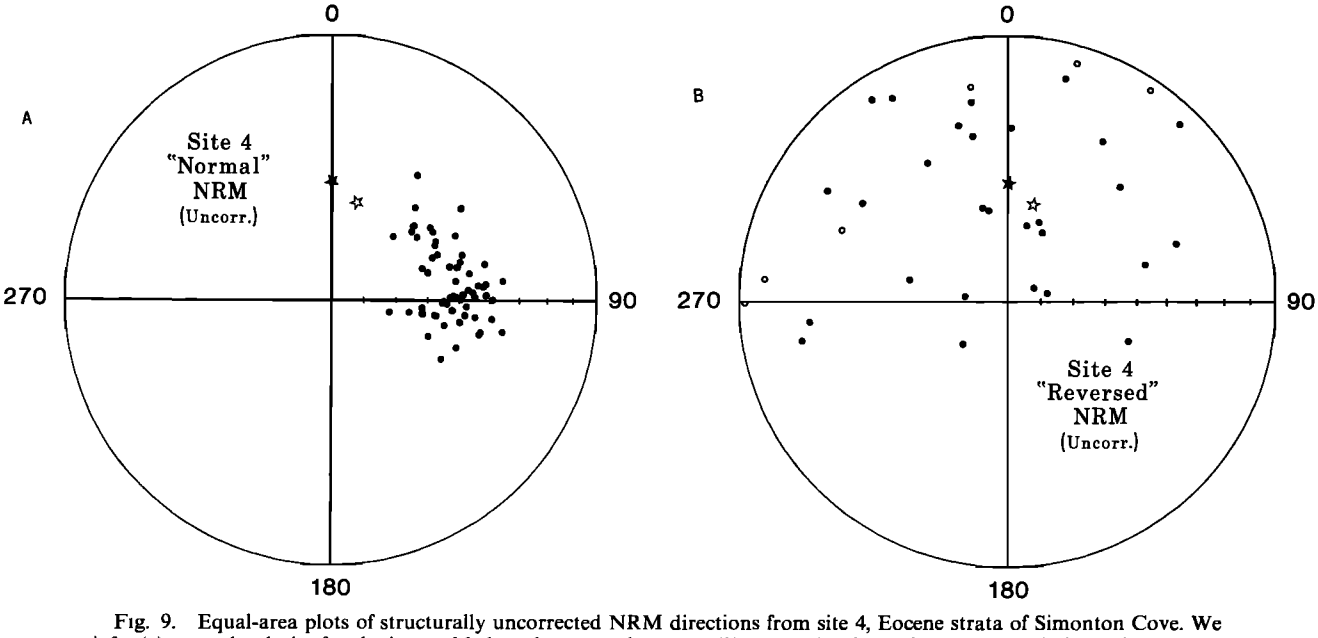


Fig. 9. Equal-area plots of structurally uncorrected NRM directions from site 4, Eocene strata of Simonton Cove. We infer (*a*) normal polarity for the interval below the covered zone or (*b*) reversed polarity for the interval above the covered zone from the sample distributions. The solid and open symbols denote data on the lower and upper hemispheres, respectively. The solid star shows the present axial dipole direction, and the open star shows the present (1980) field direction for the collecting site.

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11557 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

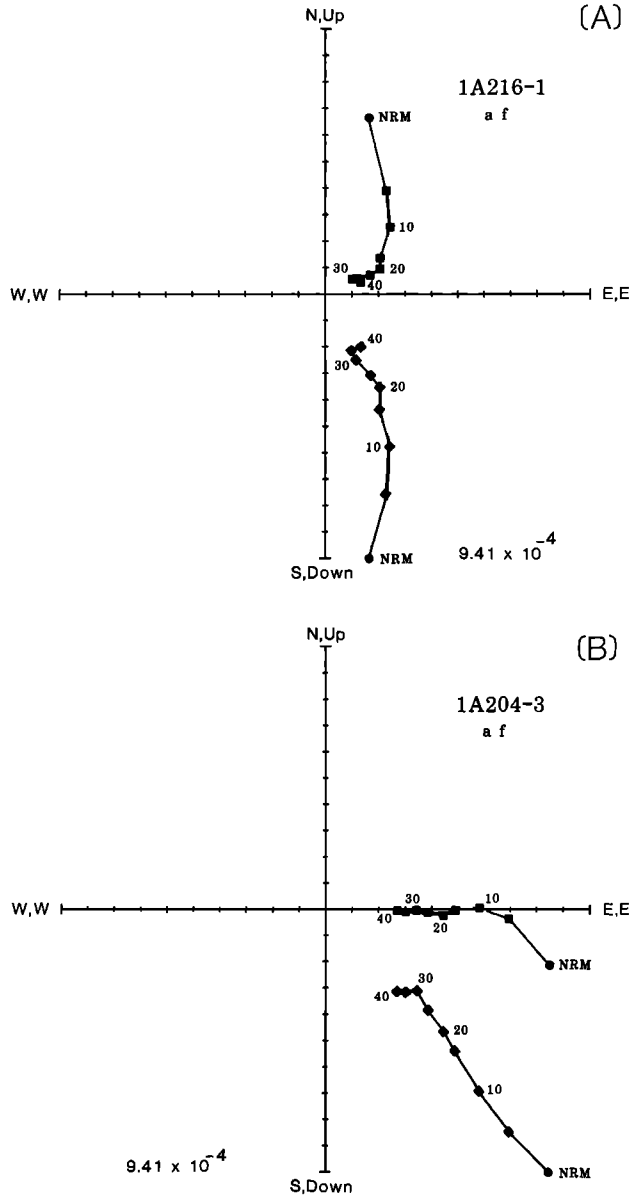


Fig. 10. Orthogonal vector component diagrams of alternating field demagnetization of two samples from Eocene strata at Simonton Cove. Squares denote the projection of the horizontal components, and diamonds denote the projection of the vertical components. Intensity values at base of diagrams indicate length of component axes in units of amperes per meter: (*a*) sample 1A216-1, showing removal of a present field component at indicated peak fields (in milliteslas), producing a linear descent toward the origin and (*b*) sample 1A204-3, showing removal of a component with moderately normal inclination and southeasterly declination, producing a stable end point.

nation and intermediate inclination (Figure 9*a*). Above the covered interval the weaker intensity samples are much more dispersed and nearer the present field direction (Figure 9*b*). The characteristics of the samples above and below the covered interval at Simonton Cove suggest different polarities for the groups or possibly different secondary magnetizations; therefore the samples were analyzed separately.

NORMAL POLARITY SAMPLE RESULTS

Specimens from each core in the stratigraphically lower, more strongly magnetized subset from site 4 responded well to stepwise af demagnetization. This cleaning technique isolated the presumed primary component of these samples; pilot thermal demagnetizations produced no improvement over the af technique. Vector component diagrams demonstrate that af demagnetization routinely produces horizontal and vertical components that linearly trend toward the origin (Figure 10). The af demagnetization of sample 1A216-1 removed a steeply inclined, northerly declined present field component at peak field strengths of 200 Oe, producing a linear descent towards the origin of the diagram. An easterly declined, shallow, secondary component was removed from sample 1A204-3, creating a stable end point direction but not a linear segment through the origin. Despite the different secondary magnetizations, the resulting demagnetized components from this subsite are closely comparable.

After stepwise af cleaning, the results from 57 of 62 original cores from the stratigraphically lower subsite were averaged. Of these, 49 samples have vector component diagrams with linear segments that trend toward the origin during some part of their demagnetization path. The remaining eight samples show good cleaning behavior and stable magnetic directions but do not produce linear segments on vector component diagrams. The five samples rejected possess exceptionally strong present field overprints. Samples from this subsite are evenly split in their direction of movement in response to af demagnetization and show both of the secondary magnetizations, depicted in the vector component diagrams of Figure 10. An equal-area plot of sample means for structurally uncorrected and corrected data shows that demagnetization shifts the mean direction away from the present field direction by a few degrees (Figure 11). This behavior is typical of turbiditic rocks from relatively fresh shoreline exposures where viscous remanence is the most important secondary magnetization [*Champion et al.*, 1984].

A fold test internal to these stratigraphically lower, normal polarity samples is insignificant, owing to limited structural

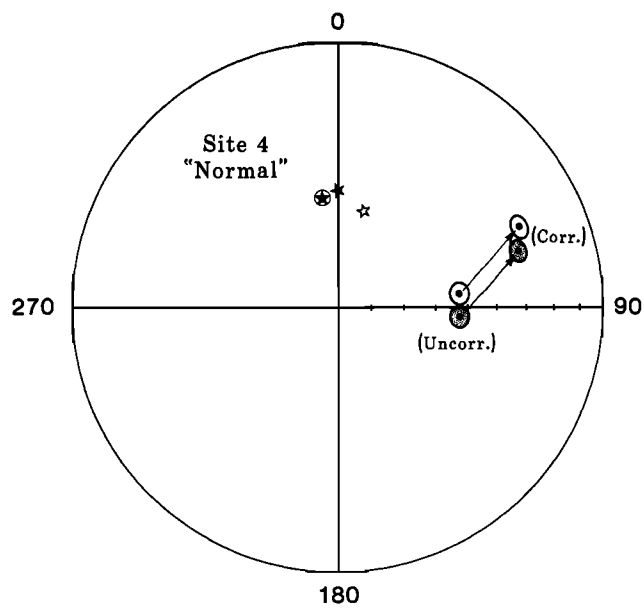


Fig. 11. Equal-area plot of mean directions of magnetization with 95% confidence circles from the normal subset of samples from site 4, Simonton Cove. Stippled confidence circles represent data cleaned by alternating field demagnetization. The solid star shows the present axial dipole direction, and the open star shows the present (1980) field direction for the collecting site. The solid star within the circle denotes the expected Eocene field direction for normal polarity at the position of San Miguel Island, assuming contiguity with North American craton.

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11557 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

diversity in the section. Structural correction of the paleomagnetic data raises the k from 35.1 to 39.8 and lowers the α_{95} from 3.2° to 3.0°, but considering the number of samples, these improvements are below the 95% confidence level (Table 2). The demagnetized and structurally corrected mean direction of magnetization for this subsite shows an inclination of 30.1° and a declination of 72.4°, which is shallower and rotated clockwise from the expected Eocene normal polarity direction for San Miguel Island. Confidence tests of this result are discussed below.

REVERSED POLARITY SAMPLE RESULTS

Specimens from each core of the stratigraphically higher, weakly magnetized subsite from site 4 were partially cleaned after stepwise af demagnetization in a manner similar to that for samples from site 3 at Cuyler Harbor. The magnetization vectors moved along two-component paths toward a reversed polarity direction with westerly declination, but they became erratic before reaching a stable direction. Alternatively, stepwise thermal demagnetization on second specimens from all 34 cores from this subsite better isolated a high blocking temperature component (Figure 12). In four specimens, thermal demagnetization produces stable directions of magnetization.

Results from 33 of 34 original cores were averaged together for this subsite. The sole rejected sample maintained an anomalous direction of magnetization. Following the procedure outlined for site 3, the mean direction of magnetization was calculated from the distribution of poles normal to the two-component planes produced by thermal demagnetization. The angular path lengths of thermal cleaning along these planes varied from 18° to 183°, and the mean angular path length is 56°. An equal-area plot of the intersection of these two-component planes, compared with the structurally corrected NRM directions, shows that the path lengths were about 1/3 of the angular distance between these points (Figure 13). The intersection of these planes is well constrained, particularly in inclination. The four samples that were thermally cleaned showed primary directions of magnetization and corroborate the mean direction (Figure 13). Using method 2 of *Kirschvink* [1980], we combined the plane and the direction information

TABLE 2. Paleomagnetic Data After af and Thermal Demagnetization, Both Structurally Uncorrected and Corrected, Taken From Sites 3 and 4 in Eocene Strata of San Miguel Island

Site	Data	N/N_0	R	κ	α_{95}	I	D
3	corr	19/21	(planes)	37.1	8.2	−29.4	221.3
4″R″	corr	29/34	(planes)	15.8	10.1	−33.5	244.1
	corr	4/34	3.9227	38.8	14.9	−30.3	247.2
Method 2		33/34	···	16.2	8.5	−33.9	244.7
4″N″	uncorr	57/62	55.40365	35.1	3.2	52.6	94.4
	corr		55.59216	39.8	3.0	30.1	72.4
Outcrop areas		2/2	1.99568	231.4	16.5	32.1	68.6
Independent cores (method 2)		90/96	(planes)	18.0	3.8	31.9	71.0
Independent orientation		29	···	···	6.7	···	···

Data sets, three structurally corrected (corr) and one uncorrected (uncorr) for two outcrop areas. Column-heading descriptions as in Table 1. Sites 3 and 4″R″ produce mean directions from analysis of the intersection of planes drawn along incomplete cleaning paths. These directions do not have uncorrected results. The "planes" solutions do not predict a value for R and produce two α_{95} values, the rms value of which appears in this table. "Method 2" in the site column refers to a method proposed by *Kirschvink* [1980] that combines "planes" data and clean directions of magnetization.

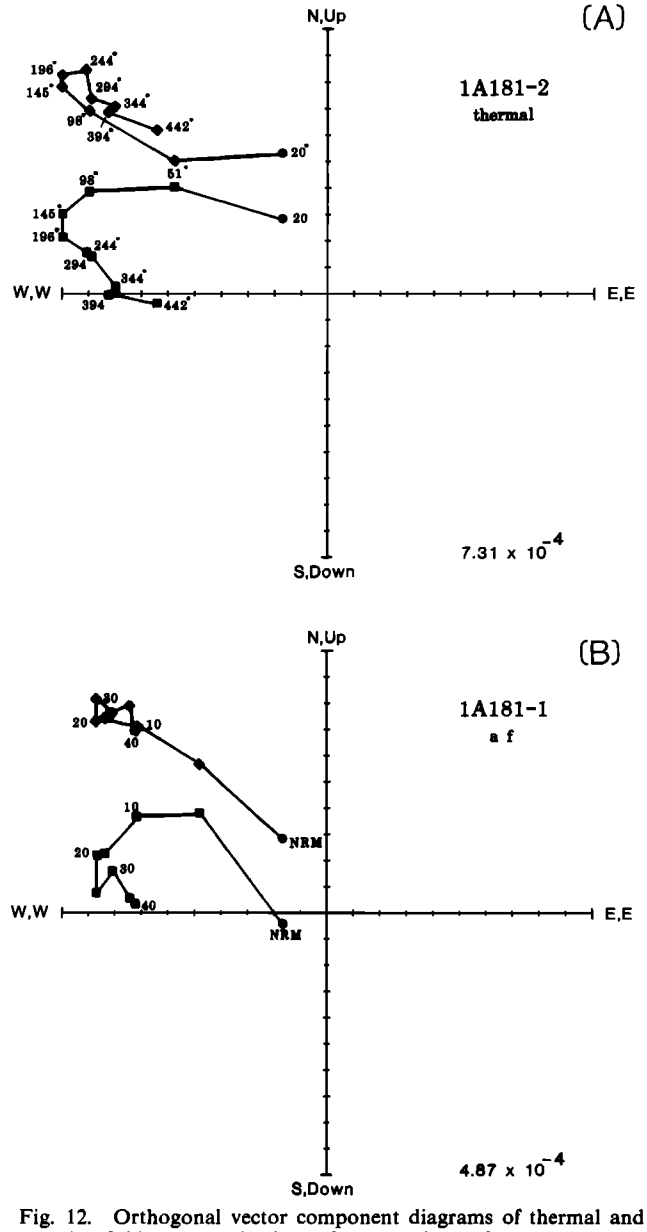


Fig. 12. Orthogonal vector component diagrams of thermal and alternating field demagnetizations of two specimens from core 1A181, Eocene strata of Simonton Cove. Squares denote the projection of the horizontal components, and diamonds denote the projection of the vertical components. Intensity values at base of diagrams indicate length of component axes in units of amperes per meter: (*a*) specimen 2, showing thermal demagnetization behavior at indicated temperatures and (*b*) specimen 1, showing alternating field demagnetization behavior at indicated peak fields (in milliteslas). Thermal cleaning better defines the directional components within the core sample.

for the reversed subsite. The larger of the two dispersion parameters from Bingham statistics is 16.2, and the 95% confidence limit of the mean direction is 8.5°, calculated from the rms of the two orthogonal α_{95} from Bingham analysis. The mean direction of magnetization has an inclination of −33.9° and a declination of 244.7°, shallower and rotated clockwise from the expected Eocene reversed polarity direction for San Miguel Island.

The origin and nature of the secondary magnetization affecting the reversed subsite at Simonton Cove and site 3 at Cuyler Harbor can be understood. It is unlike the viscous secondary magnetic effects that usually affect these fresh turbi-

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11557 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

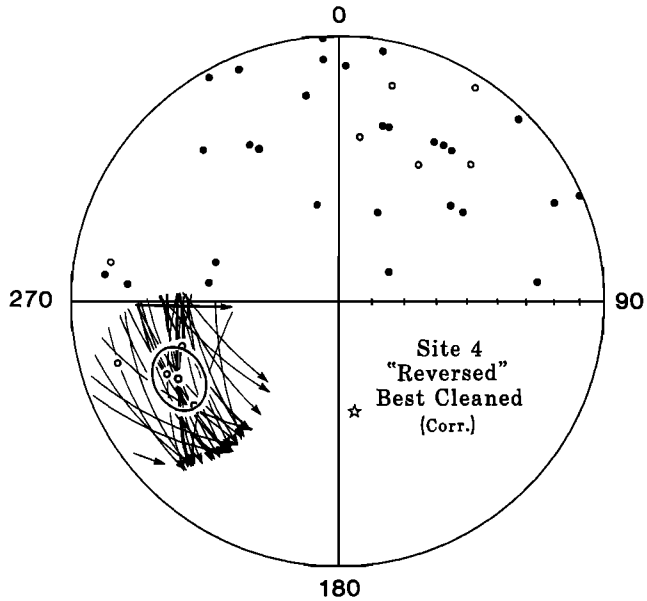


Fig. 13. Equal-area plot of structurally corrected NRM field directions from the reversed subset of samples from site 4, Eocene strata of Simonton Cove. Arcs with arrows denote the trace of great circle thermal-cleaning paths from these NRM directions and the sense of the cleaning. The four reversed field directions (light open circles) plotted within the area of cleaning arcs represent samples thermally cleaned of secondary magnetizations. Mean direction of magnetization with 95% confidence circle is calculated from the distribution of poles to planes, combined with the four cleaned directions, by a method described by *Kirschvink* [1980]. Open star denotes the expected Eocene field direction for reversed polarity at the position of San Miguel Island, assuming contiguity with North American craton.

dite sediments and evidently cannot be completely removed by using routine af demagnetization. An equal-area plot of uncorrected low-temperature difference vectors, calculated between successive demagnetization steps from the reversed samples, shows them to be far from the present field or axial dipole direction (Figure 14). Instead, they have easterly to southeasterly declinations and intermediate normal inclinations. These directions are close to those obtained by *Kamerling and Luyendyk* [1985] from basaltic dikes that intrude the upper part of the section at Simonton Cove. These Miocene volcanic rocks are associated with those from the early Miocene volcanic activity at Harris Point, which are also close to the badly overprinted samples at site 3. Evidently, the volcanic rocks heated the underlying Eocene sedimentary rocks and imparted a secondary thermal magnetization. This effect dissipated stratigraphically downward, such that the lower part of the Simonton Cove section, below the covered interval, is less affected. Alternatively, the heat of the overlying volcanic deposits may have caused circulating ground water to precipitate authigenic iron oxides, creating a Miocene chemical remanence in the underlying sediments.

The normal and reversed samples at Simonton Cove, though cleaned by different techniques due to differing secondary magnetizations, compare favorably in a reversal test (Figure 15). There is no significant difference in their mean directions at the 95% confidence level [*McFadden and Jones*, 1981]. The mean direction for site 4 was calculated using method 2, combining the directions and planes information from 90 of 96 original cores [*Kirschvink*, 1980], and yielded an inclination of 31.9° and a declination of 71.0°, with an rms α_{95} of 3.8°. A better figure for a 95% confidence limit, calculated from $N = 29$ independent structural attitudes taken and using the $k = 18.0$ value, gives a value of $\alpha_{95} = 6.7°$. The demagnetized and structurally corrected mean direction for site 4 is 23.9° ± 5.7° shallower in inclination and rotated 79.4° ± 6.8° clockwise from the expected Eocene field direction. The mean inclination value corresponds to a paleolatitude of 17.3°N. The inclination anomaly corresponds to a poleward motion with respect to North America of 19.1°.

The mean directions of magnetization from sites 3 and 4 are rotated clockwise from the expected Eocene direction by about 50° and 80°, respectively. This difference may be an artifact of the planes solutions used in part for both sites and of the particular orientations of primary and overprinting directions. This artifact causes more of the error in the mean directions to be in the declination. An equally valid possibility is that the Neogene dextral shear process, which brought the northern Channel Islands to their present position from the San Diego area [*Luyendyk et al.*, 1985], broke them into blocks having individual rotation histories. This explanation would also address the curious situation in which the Upper Cretaceous strata at sites 1 and 2 are rotated 30° less than the Eocene strata at site 4, just 4 km up the coast. The mapped extension of the sinistral Santa Rosa fault passes just south of San Miguel Island, and the strata at Point Bennett would be vulnerable to counterclockwise rotations associated with late Neogene displacement on that fault [*Weaver et al.*, 1969]. Alternatively, it is possible, but improbable, that in Cretaceous and/or Paleocene time the entire region underwent counterclockwise rotation.

DISPLACEMENT HISTORY OF THE BAJA-BORDERLAND COMPOSITE TERRANE

Stratigraphic relations among the Nicolas and adjoining terranes provide a time frame for terrane amalgamation [*Howell et al.*, 1985]. These relations suggest that by Late Cretaceous time the Nicolas, Malibu, Cortez, and Santa Ana terranes were joined to form a major allochthon, the Baja-

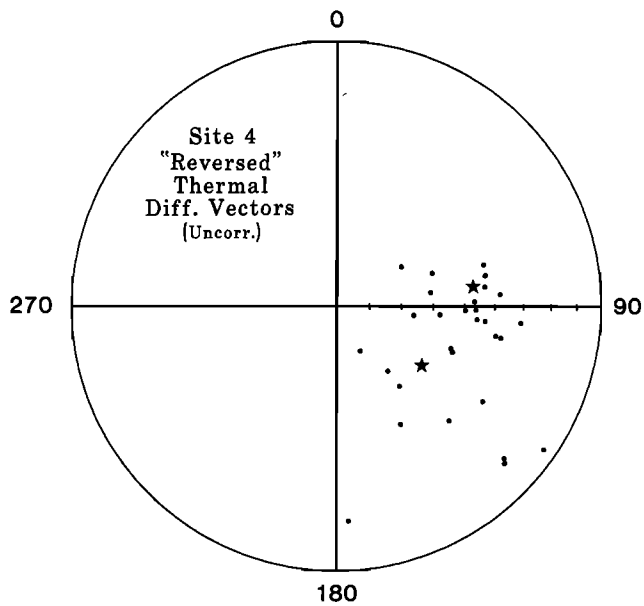


Fig. 14. Equal-area plot of structurally uncorrected low-temperature difference vectors from the reversed subset of samples from site 4, Eocene strata of Simonton Cove. Solid stars denote directions from two sites in the northeastern part of the Simonton Cove section in Miocene volcanic dikes (MCA and MCB, from *Kamerling and Luyendyk* [1985]), believed to be responsible for the thermal overprints on the samples with reversed polarity.

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11557 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Borderland allochthon of *Howell et al.* [1986]. The stratigraphy, however, does not indicate the degree of allochthoneity of these terranes. Paleomagnetic data do suggest poleward motions of the Baja California peninsula that were not predicted by conventional geologic correlations [*Hagstrum et al.*, 1985; *Fry et al.*, 1985]. Geologic relations indicate a late Miocene and younger opening of the Gulf of California, displacing the terranes west of the gulf toward the north a distance of 260 km [*Larson et al.*, 1968]. Middle to late Miocene sedimentary and volcanic lithofacies across the gulf constrain any additional slip to pre-15 Ma [*Gastil et al.*, 1981; *Hausback*, 1984]. Mesozoic volcanic arc rocks crop out both on the Baja California peninsula and in western Sonora. Published tectonic reconstructions either palinspastically align the arc strata into a single belt [*Hamilton*, 1969; *Gastil et al.*, 1972; *Silver et al.*, 1979] or propose the collapse of paired volcanic arcs and the destruction of a marginal basin separating them [*Rangin*, 1978; *Gastil et al.*, 1981].

Early paleomagnetic data from Peninsular Range batholithic rocks pointed to the possibility of an original latitude for southern California that was south of positions realized by the simple closing of the Gulf of California [*Teissere and Beck*, 1973]. Subsequent studies on Cretaceous sedimentary rocks in the San Diego area [*Marshall and McNaboe*, 1979], in the Santa Ana Mountains [*Fry et al.*, 1985], and on rocks of the Vizcaino Peninsula [*Patterson*, 1984] support these low late Mesozoic latitudes for peninsular California. *Hagstrum et al.* [1985], working with paleomagnetic data from both plutonic and sedimentary rocks, position the peninsula in Cretaceous time at or near 30°N, south of the expected Cretaceous latitude of 42°N.

The paleomagnetic results from our study corroborate this

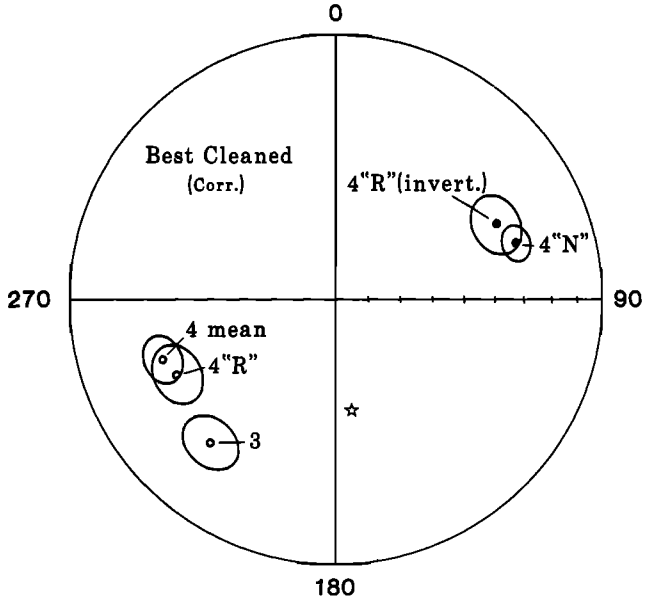


Fig. 15. Equal-area plot of structurally corrected best cleaned field directions with 95% confidence circles from sites 3 and 4, Eocene strata of San Miguel Island. Numbers relate to sampling sites, with "N" and "R" denoting the normal and reversed subsets of samples from Simonton Cove. Open star denotes the expected Eocene field direction for reversed polarity at the position of San Miguel Island, assuming contiguity with North American craton. The reversed direction, shown as 4"R", has also been inverted to a south seeking direction for direct comparison with 4"N." They agree at the 95% confidence level, but the mean directions from sites 3 and 4 do not agree. This difference is due either to different structural rotations or to a poor declination determination for site 3.

TABLE 3. Summary Table of Paleomagnetic Data From Cretaceous and Eocene Rocks of San Miguel Island

	Site		
	1, 2	3	4
Age	91–65 Ma	52–42 Ma	55–52 Ma
I_{obs}	44	−29	32
D_{obs}	24	221	71
P_{lat}	68	49	25
P_{long}	345	346	325
Lat_{paleo}	26	16	17
ΔL	19	21	19
ΔR	50	50	79

Age is biostratigraphic age range at each site, I_{obs} and D_{obs} are the observed inclination and declination, P_{lat} and P_{long} are the VGP latitude and longitude calculated from the observed inclination and declination and the site location, Lat_{paleo} is the paleolatitude calculated from the observed inclination, ΔL is the poleward shift, and ΔR is the amount of clockwise rotation calculated from the site VGP as compared to the pole for the craton of the same age.

southerly location (Table 3). The aforementioned results are plotted on a latitude versus time diagram for the various terrane elements of the Baja-Borderland allochthon [Figure 16]. The locus of the latitudinal history of the North American craton, plotted at the present position of San Diego, provides a frame of reference for possible motions [*Irving and Irving*, 1982]. The latitudinal history for the craton at San Diego requires northward movement in Jurassic and Early Cretaceous time. During this interval the apparent polar wander path approaches the Bering Strait from Siberia. A static and relatively high latitude for the San Diego area exists for most of the Cretaceous, and this was followed by a 20-m.y. episode during the Paleogene of southward motion that ended before Oligocene time (Figure 16).

During this late Mesozoic–early Cenozoic time interval the Baja-Borderland allochthon seemingly mimics North American cratonal motion but at more southerly latitudes than expected. *Hagstrum et al.* [1985] show that Upper Jurassic–Lower Cretaceous pillow lavas of the Eugenia Formation of the Vizcaino Peninsula of northwest Baja California Sur have latitudes of 14°, although the hemisphere of origin is in doubt. The Cretaceous latitude obtained for the Santa Ana–Cortez portions of the Baja California peninsula and for the Vizcaino Peninsula requires that they have undergone little motion relative to the North American craton since Early Cretaceous time. Considering the high rates of northward motion required to bring the Vizcaino Peninsula to a position where Early Cretaceous Valle Formation sediments can be deposited on it and the main axis of the Baja peninsula, without relative displacement, the northern hemisphere choice of origin for the Eugenia Formation is more reasonable. The Eugenia Cretaceous movement proposed by *Hagstrum et al.* [1985] thus shows how closely the Baja California peninsula movement has paralleled cratonal trajectories.

The paleomagnetic results from Late Cretaceous sediments on San Miguel Island indicate a paleolatitude of 26°N, just south of the paleomagnetically determined latitude of the San Diego area [*Hagstrum et al.*, 1985]. Contrasting Late Cretaceous sediments in the San Diego area and on San Miguel Island indicate that the San Miguel Cretaceous sediments were deposited at least 100 km south of the San Diego area and south of the Valle de las Palmas area in order to receive Alisitos arc detritus [*Bartling*, 1981]. But juxtaposition of the northern Channel Islands and San Diego areas is required by Eocene time to explain the distribution of lithofacies and

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11557 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

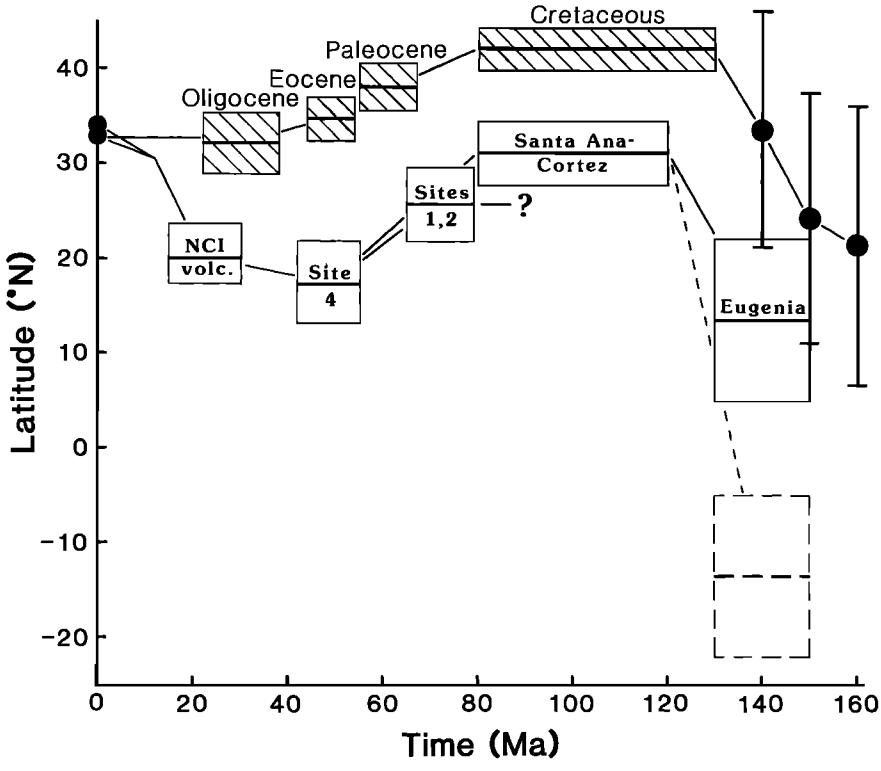


Fig. 16. Diagram showing paleolatitudes versus time for cratonal North America at the hypothetical position of San Diego and terranes of the Baja-Borderland allochthon. Diagonally ruled boxes represent calculations derived from paleomagnetic data for cratonal North America. The data for Tertiary time are from *Diehl et al.* [1983]; for Cretaceous time we depict the mean of entries 39, 40, 42, 43, 44, and 45 of Table I of *Irving and Irving* [1982]; and for late Jurassic time we use 30 Ma mean paleopoles calculated for 140, 150, and 160 Ma by *Irving and Irving* [1982]. Open boxes are observed data from indicated rocks presented by this study, studies on northern Channel Island volcanic rocks by *Kamerling and Luyendyk* [1985], or from *Hagstrum et al.* [1985]. Vertical dimensions of boxes are within 95% confidence ranges, and horizontal dimensions indicate ranges in estimated ages for the sampled strata.

Poway clast assemblages in both areas [*Howell and Link*, 1979; *Kies and Abbott*, 1983]. *Bartling* [1981] hypothesizes an early Tertiary episode of right slip between the northern Channel Islands and the San Diego area. Because of the Cretaceous and Eocene correlations between the Nicolas and Peninsular Range terranes, the paleomagnetic results from San Miguel Island precisely locate the northern Baja California peninsula and the north part of the Nicolas terrane at 17°N latitude in Eocene time. A position at this southerly latitude as recently as the Eocene requires that the Baja-Borderland allochthon continues to generally mimic the cratonal motion throughout the Paleogene, thus suggesting that the allochton was still attached to the craton at that time.

The precise trajectory of the Baja-Borderland allochthon is poorly constrained in post-Eocene time. We do know that a fundamental rearrangement of oceanic plate motions occured in mid-Eocene [*Engebretson*, 1983], and it seems likely that the allochthon became detached from North America at that time. Analysis of relative motions suggests that the Farallon plate may have exerted right shear that paralleled the North American coast at these southerly latitudes [*Beck*, 1986]. Paleomagnetic results from early and mid-Miocene volcanic rocks in the northern parts of the Baja-Borderland allochthon show the terranes had reached a latitude of 20°N by that time [*Kamerling and Luyendyk*, 1985]. Other paleomagnetic work by J. T. Hagstrum et al. (unpublished manuscript, 1986) suggests that the Baja California peninsula had reached its pre–Gulf of California position at or before earliest Miocene time. Geologic correlations [*Hausback*, 1984; *Gastil et al.*, 1979] tie the allochthon to its pre–Gulf of California latitude in mid-Miocene time.

Comparison of Displacement Histories of Baja-Borderland and Santa Lucia–Orocopia Allochthons

The trajectory of the Baja-Borderland allochthon is different from the trajectory of the other allochthonous terranes in southern California, especially the Santa Lucia–Orocopia allochthon of *Vedder et al.* [1983] [*McWilliams and Howell*, 1982; *Champion et al.*, 1984]. The relative displacement histories of those two allochthons are shown in Figure 17, along with two dotted paths of cratonal movement, calculated for the position of San Francisco and San Diego. The latitudinal paths are shown schematically, with 5°-wide bands of latitude and errors of ±10 m.y. in age. The latitudinal paths for both allocthons are for the northern parts of these elongate allochthons (the Baja-Borderland allochthon is about 1400 km long).

A summary of the paleomagnetic data shows that in Late Jurassic to Early Cretaceous time the oceanic island arc Stanley Mountain terrane experienced northward motion, amalgamating with the Salinia and Tujunga terranes in latest Cretaceous at about 20°N latitude. Then, most likely riding with the Kula or some other unknown plate, the now formed Santa Lucia–Orocopia allochthon moved rapidly northward until Eocene time, when it accreted and partly overrode the North American craton [*Vedder et al.*, 1983]. In the same time interval the Santa Ana and Cortez terranes of the Baja-Borderland

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11557 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

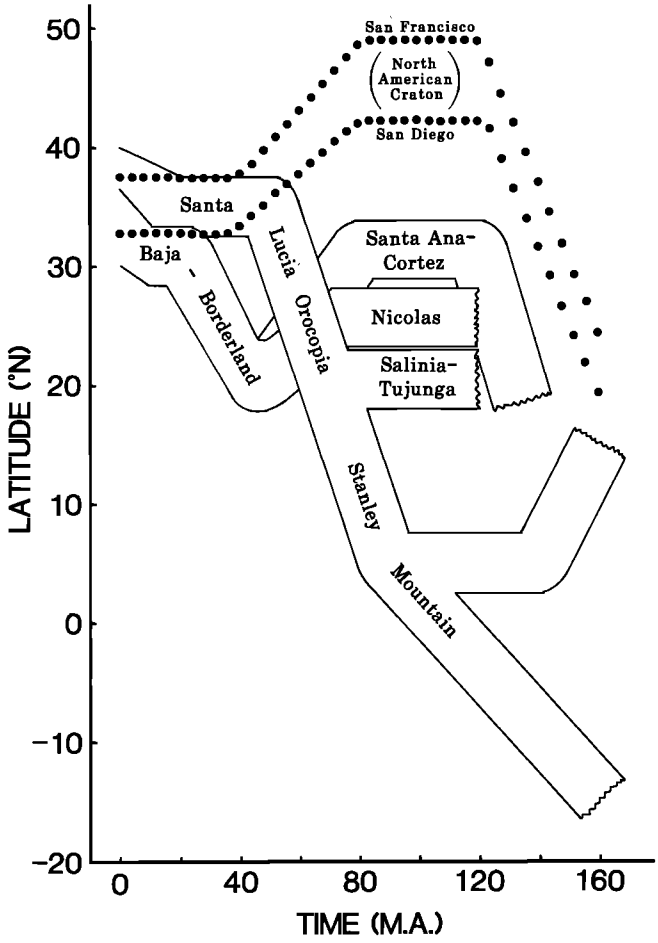


Fig. 17. Relative late Mesozoic and Tertiary displacement histories for the Baja-Borderland and Santa Lucia–Orocopia allochthons. Dotted paths represent craton trajectories calculated for the positions of San Francisco and San Diego. Paths of motion are shown schematically with a latitude error of 5° and a time error of ±10 m.y. Where trajectories of terranes or allochthons cross, the most seaward trajectory is shown overlapping those inland.

allochthon were moving along with the craton. The Nicolas terrane, caught in a dextral shear between the northward moving Santa Lucia–Orocopia allochthon and the pluton-stitched terranes of the Baja-Borderland allochthon, did not move as far (Figure 17). In Eocene time the Baja-Borderland allochthon was detached from the craton and ceased mimicking cratonal trajectories. The renewed northward motion, presumably accompanying the Farallon plate, resulted in its reaching northern Sonora latitudes (30°N) by about 25 Ma, when the rate of poleward motion seems to have slowed. To its north the Santa Lucia–Orocopia allochthon remained fixed to North America until about 24 Ma, when the San Andreas fault system began dismembering the assemblage by slicing through the allochthon moving segments northward to the region now occupied by the central California Coast Ranges.

CONCLUSION

Paleomagnetic data from the terranes of southwestern North America evince a protracted history of lateral transport along the continental margin. This slivering has repositioned Jurassic-Cretaceous volcanic arc strata, thus rendering suspect the simple palinspastic models that (1) align the Sierra Nevada, Salinian, Peninsula Range, and Baja Peninsula terranes, in that order north to south, into a single linear volcanic arc, and (2) limit the southward relocation of the Baja California peninsula to the closing of the Gulf of California.

The paleomagnetic and stratigraphic data from San Miguel Island suggest that several terranes accreted to the craton at the position of southern Mexico and moved with the craton during Cretaceous and Paleocene time before dispersal processes resulted in coastwise-poleward transport. A summary of all available paleomagnetic data from southern California and the Baja peninsula requires plate interactions involving the North America, Kula, Farallon, and possible other now subducted plates [*Tarduno et al.*, 1986] that resulted in a complicated history involving amalgamations, accretions, oblique transpression, and dispersion. This slivering and transport of pieces of western North America is active today along the San Andreas fault system.

Acknowledgments. Thank yous are in order to the National Park Service for granting permission to perform paleomagnetic sampling in the Channel Islands National Monument and to the U.S. Navy for providing transportation to and from San Miguel Island. Guidance to key outcrops was provided by Bill Bartling and Ron Kies, then of San Diego State University. Able field assistance was supplied by Tom Jones. Reviews of the manuscript by Jon Hagstrum, Clark Blake, Michael McWilliams, and an unknown reviewer clarified and strengthened the text. Thanks are also in order to Jon Hagstrum for allowing access to and mention of unpublished data.

REFERENCES

Bartling, W. A., Cretaceous sedimentology of San Miguel Island and its relationship to mainland outcrops, M.A. thesis, 262 pp., San Diego State Univ., San Diego, Calif., 1981.

Beck, M. E., Jr., Discordant paleomagnetic pole positions as evidence of regional shear in the Western Cordillera of North America, *Am. J. Sci.*, *276*, 694–712, 1976.

Beck, M. E., Jr., Paleomagnetic record of plate-margin tectonic processes along the western edge of North America, *J. Geophys. Res.*, *85*, 7115–7131, 1980.

Beck, M. E., Jr., Model for late Mesozoic–early Tertiary tectonics of coastal California and western Mexico and speculations on the origin of the San Andreas fault, *Tectonics*, *5*, 49–64, 1986.

Beck, M. E., Jr., R. F. Burmester, D. C. Engebretson, and R. Schoonover, Northward translation of Mesozoic batholiths, western North America: Paleomagnetic evidence and tectonic significance, *Geofis. Int.*, *20*(3), 143–162, 1981.

Blake, M. C., D. G. Howell, and D. L. Jones, Preliminary tectonostratigraphic terrane map of California, *U.S. Geol. Surv. Open File Rep.*, *82-593*, 10 pp., 1982.

Bouma, A. H., *Sedimentation of Some Flysch Deposits: A Graphic Approach to Facies Interpretations*, 168 pp., Elsevier Science, New York, 1962.

Champion, D. E., D. G. Howell, and C. S. Gromme, Paleomagnetic and geologic data indicating 2500 km of northward displacement for the Salinian and related terranes, California, *J. Geophys. Res.*, *89*, 7736–7752, 1984.

Diehl, J. F., M. E. Beck, Jr., S. Beske-Diehl, D. Jacobson, and B. C. Hearn, Jr., Paleomagnetism of the Late Cretaceous–early Tertiary north central Montana alkalic province, *J. Geophys. Res.*, *88*, 10,593–10,609, 1983.

Engebretson, D. C., Relative motions between the North American, Kula, Farallon, and Pacific plates from 145 my to the present, Ph.D. thesis, 211 pp., Stanford Univ., Stanford, Calif., 1983.

Filmer, P. E., and J. L. Kirschvink, Magnetostratigraphy of the Upper Cretaceous Rosario Formation, northwestern Baja California, Mexico (abstract), *Geol. Soc. Am. Abstr. Programs*, *18*, 106, 1986.

Fry, J. G., D. J. Bottjer, and S. P. Lund, Magnetostratigraphy of displaced Upper Cretaceous strata in southern California, *Geology*, *13*, 648–651, 1985.

Gastil, G., D. Krummenacher, and J. Minch, The record of Cenozoic volcanism around the Gulf of California, *Geol. Soc. Am. Bull.*, *90*, 839–857, 1979.

Gastil, G., G. J. Morgan, and D. Krummenacher, The tectonic history of peninsular California and adjacent Mexico, in *The Geotectonic Development of California*, edited by W. G. Ernst, pp. 2–28, Prentice-Hall, Englewood Cliffs, N. J., 1981.

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11557 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Gastil, R. G., R. P. Phillips, and R. Rodriquez-Torres, The reconstruction of Mesozoic California, *Int. Geol. Congr. Rep. Sess. 24th, Sect. 3*, 217–229, 1972.

Hagstrum, J. T., M. O. McWilliams, D. G. Howell, and C. S. Gromme, Mesozoic paleomagnetism and northward translation of the Baja California peninsula, *Geol. Soc. Am. Bull., 96*, 1077–1090, 1985.

Hamilton, W., Mesozoic California and the underflow of Pacific mantle, *Geol. Soc. Am. Bull., 80*, 2409–2430, 1969.

Hausback, B. P., Cenozoic volcanic and tectonic evolution of Baja California Sur, Mexico, in *Geology of the Baja California Peninsula*, edited by V. A. Frizzell Jr., pp. 219–236, Society of Economic Paleontologists and Mineralogists, Pacific Coast Section, Los Angeles, Calif., 1984.

Howell, D. G., and M. H. Link, Eocene conglomerate sedimentology and basin analysis, San Diego and the southern California borderland, *J. Sediment. Petrol., 49*, 517–540, 1979.

Howell, D. G., C. J. Stuart, J. P. Platt, and D. J. Hill, Possible strike-slip faulting in the southern California borderland, *Geology, 2*, 93–98, 1974.

Howell, D. G., J. D. Gibson, G. S. Fuis, J. H. Knapp, G. B. Haxel, B. R. Keller, L. T. Silver, and J. G. Vedder, Ocean-continent transect C-3 (southern California borderland to Rio Grande rift), 3 sheets, 23 pp., Transl. Program, Geol. Soc. Am., Boulder, Colo., 1985.

Howell, D. G., D. E. Champion, and J. G. Vedder, Terrane accretion, crustal kinematics and basin evolution, southern California, *Rubey Vol. 2*, in press, Prentice Hall, Englewood Cliffs, N. J., 1986.

Irving, E., and G. A. Irving, Apparent polar wander paths, Carboniferous through Cenozoic and the assembly of Gondwana, *Geophys. Surv., 5*, 141–188, 1982.

Kamerling, M. J., and B. P. Luyendyk, Paleomagnetism and Neogene tectonics of the northern Channel Islands, California, *J. Geophys. Res., 90*, 12,485–12,502, 1985.

Kies, R. P., Paleogene sedimentology, lithostratigraphic correlations and paleogeography, San Miguel Island, Santa Cruz Island, and San Diego, California, M.A. thesis, 577 pp., San Diego State Univ., San Diego, Calif., 1982.

Kies, R. P., and P. L. Abbott, Rhyolite clast populations and tectonics in the California borderland, *J. Sediment Petrol., 53*, 461–475, 1983.

Kirschvink, J. L., The least-squares line and plane and analysis of palaomagnetic data, *Geophys. J. R. Astron. Soc., 62*, 699–718, 1980.

Larson, R. L., H. W. Menard, and S. M. Smith, Gulf of California: A result of ocean-floor spreading and transform faulting, *Science, 161*, 781–784, 1968.

Luyendyk, B. P., M. J. Kamerling, R. R. Terres, and J. S. Hornafius, Simple shear of southern California during Neogene time suggested by paleomagnetic declinations, *J. Geophys. Res., 90*, 12,454–12,466, 1985.

Marshall, M., and G. J. McNaboe, Preliminary paleomagnetic results from the Cretaceous Point Loma Formation, in *Geological Excursions in the Southern California Area*, edited by P. L. Abbott, p. 183, Department of Geological Sciences, San Diego State University, San Diego, Calif., 1979.

McFadden, P. L., and D. L. Jones, The fold test in paleomagnetism, *Geophys. J. R. Astron. Soc., 67*, 53–58, 1981.

McWilliams, M. O., and D. G. Howell, Exotic terranes of western California, *Nature, 297*, 215–217, 1982.

Patterson, D. L., Paleomagnetism of the Valle fomation and the Late Cretaceous paleogeography of the Vizcaino basin, Baja California, Mexico, in *Geology of the Baja California Peninsula*, edited by V. A. Frizzell, Jr., pp. 173–182, Society of Economic Paleontologists and Mineralogists, Pacific Coast Section, Los Angeles, Calif., 1984.

Rangin, C., Speculative model of Mesozoic geodynamics, central Baja California to northeastern Sonora (Mexico), in *Mesozoic Paleogeography of the Western United States*, edited by D. G. Howell and K. A. McDougall, pp. 85–106, Society of Economic Paleontologists and Mineralogists, Pacific Coast Paleogeography Symposium 2, Sacramento, Calif., 1978.

Silver, L. T., H. P. Taylor, Jr., and B. Chappell, Some petrological, geochemical and geochronological observations of the Peninsular Ranges batholith near the international border of the U.S.A. and Mexico, in *Mesozoic Crystalline Rocks, Peninsular Ranges Batholith and Pegmatite and the Point Sal Ophiolite*, edited by P. L. Abbott and V. R. Todd, pp. 83–110, Department of Geological Sciences, San Diego State University, San Diego, Calif., 1979.

Tarduno, J. A., M. McWilliams, W. V. Sliter, H. E. Cook, M. C. Blake, and I. Premoli-Silva, Southern hemisphere origin of the Cretaceous Laytonville Limestone of California, *Science, 231*, 1425–1428, 1986.

Teissere, R. F., and M. E. Beck, Jr., Divergent Cretaceous paleomagnetic pole position for the southern California batholith, U.S.A., *Earth Planet. Sci. Lett., 18*, 296–300, 1973.

Vedder, J. G., D. G. Howell, and H. McLean, Stratigraphy, sedimentation, and tectonic accretion of exotic terranes, southern Coast ranges, California, *Mem. Am. Assoc. Pet. Geol., 34*, 471–496, 1983.

Weaver, D. W., D. P. Doerner, and B. Nolf, *Geology of the Northern Channel Islands*, 200 pp., Pacific Section, American Association of Petroleum Geologists, Bakersfield, Calif., 1969.

D. E. Champion and D. G. Howell, U.S. Geological Survey, MS 18, 345 Middlefield Road, Menlo Park, CA 94025.

M. Marshall, Geology Department, San Diego State University, San Diego, CA 92180.

(Received January 3, 1986;
revised June 4, 1986;
accepted June 23, 1986.)

21562202b, 1986, B11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB091iB11p11557 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License